



1. Introduction ⁽¹⁾

[cf. Simone Fischer-Hübner]

- **Def. of privacy** [Alan Westin, Columbia University, 1967]
= the claim of individuals, groups and institutions to determine for themselves, when, how and to what extent information about them is communicated to others
- **3 dimensions of privacy:**
 - 1) **Personal privacy**
Protecting a person against undue interference (such as physical searches) and information that violates his/her moral sense
 - 2) **Territorial privacy**
Protecting a physical area surrounding a person that may not be violated without the acquiescence of the person
 - Safeguards: laws referring to trespassers search warrants
 - 3) **Informational privacy**
Deals with the gathering, compilation and selective dissemination of information

- **Basic privacy principles**
 - Lawfulness and fairness
 - Necessity of data collection and processing
 - Purpose specification and purpose binding
 - There are no "non-sensitive" data
 - Transparency
 - Data subject's right to information correction, erasure or blocking of incorrect/ illegally stored data
 - Supervision (= control by independent data protection authority) & sanctions
 - Adequate organizational and technical safeguards
- **Privacy protection can be undertaken by:**
 - Privacy and data protection **laws** promoted **by government**
 - **Self-regulation** for fair information practices by codes of conducts promoted **by businesses**
 - **Privacy-enhancing technologies (PETs)** adopted by individuals
 - **Privacy education** of consumers and IT professionals



2. Recognition of Need for Privacy Guarantees ⁽¹⁾

- By individuals

[Cran *et al.* '99]

- 99% unwilling to reveal their SSN
- 18% unwilling to reveal their... favorite TV show

- By businesses

- Online consumers worrying about revealing personal data held back \$15 billion in online revenue in 2001

- By Federal government

- Privacy Act of 1974 for Federal agencies
- Health Insurance Portability and Accountability Act of 1996 (HIPAA)



2. Recognition of Need for Privacy Guarantees (2)

- By computer **industry research** (examples)
 - **Microsoft Research**
 - The biggest research challenges:

According to Dr. Rick Rashid, Senior Vice President for Research

 - Reliability / Security / **Privacy** / Business Integrity
 - Broader: application integrity (just “integrity?”)

=> MS Trustworthy Computing Initiative
 - **Topics include:** DRM—digital rights management (incl. watermarking surviving photo editing attacks), software rights protection, intellectual property and content protection, database privacy and p.-p. data mining, anonymous e-cash, anti-spyware
- **IBM** (incl. Privacy Research Institute)
 - **Topics include:** pseudonymity for e-commerce, EPA and EPAL—enterprise privacy architecture and language, RFID privacy, p.-p. video surveillance, federated identity management (for enterprise federations), p.-p. data mining and p.-p. mining of association rules, hippocratic (p.-p.) databases, online privacy monitoring



2. Recognition of Need for Privacy Guarantees (3)

- By academic researchers (examples from the U.S.A.)
 - CMU and Privacy Technology Center
 - Latanya Sweeney (k-anonymity, SOS—Surveillance of Surveillances, genomic privacy)
 - Mike Reiter (Crowds – anonymity)
 - Purdue University – CS and CERIAS
 - Elisa Bertino (trust negotiation languages and privacy)
 - Bharat Bhargava (privacy-trust tradeoff, privacy metrics, p.-p. data dissemination, p.-p. location-based routing and services in networks)
 - Chris Clifton (p.-p. data mining)
 - Leszek Lilien (p.-p. data dissemination)
 - UIUC
 - Roy Campbell (Mist – preserving location privacy in pervasive computing)
 - Marianne Winslett (trust negotiation w/ controlled release of private credentials)
 - U. of North Carolina Charlotte
 - Xintao Wu, Yongge Wang, Yuliang Zheng (p.-p. database testing and data mining)

Why Is Privacy Important?

- Data is a corporate asset, like any other
- Corporate data is at a higher risk of theft or misuse than ever before
- Companies have obligations to protect data
 - Laws, regulations, guidelines
 - Contracts with third parties
 - Privacy policies for users of websites, other online features

Privacy and Computers

- ❖ Computers are not necessary for the invasion of privacy
- ❖ Computers embody many threats to privacy
- ❖ Computers make it faster and easier to violate privacy
- ❖ Much of our private information is just 'out there' in computers



3. Threats to Privacy (1)

[cf. Simone Fischer-Hübner]

1) Threats to privacy at application level

- Threats to collection / transmission of large quantities of personal data
 - Incl. projects for new applications on Information Highway, e.g.:
 - Health Networks / Public administration Networks
 - Research Networks / Electronic Commerce / Teleworking
 - Distance Learning / Private use
 - Example: Information infrastructure for a better healthcare
 - [cf. Danish "INFO-Society 2000"- or Bangemann-Report]
 - National and European healthcare networks for the interchange of information
 - Interchange of (standardized) electronic patient case files
 - Systems for tele-diagnosing and clinical treatment

2) Threats to privacy at communication level

- Threats to anonymity of sender / forwarder / receiver
- Threats to anonymity of service provider
- Threats to privacy of communication
 - E.g., via monitoring / logging of transactional data
 - Extraction of user profiles & its long-term storage

3) Threats to privacy at system level

- E.g., threats at system access level

4) Threats to privacy in audit trails

- **Identity theft** – the most serious crime against privacy
- **Threats** to privacy – another view
 - Aggregation and data mining
 - Poor system security
 - Government threats
 - Gov't has a lot of people's most private data
 - Taxes / homeland security / etc.
 - People's privacy vs. homeland security concerns
 - The Internet as privacy threat
 - Unencrypted e-mail / web surfing / attacks
 - Corporate rights and private business
 - Companies may collect data that U.S. gov't is *not* allowed to
 - Privacy for sale - many traps
 - "Free" is not free...
 - E.g., accepting frequent-buyer cards reduces your privacy



4. Privacy Controls

- 1) Technical privacy controls - Privacy-Enhancing Technologies (PETs)
 - a) Protecting user identities
 - b) Protecting usee identities
 - c) Protecting confidentiality & integrity of personal data
- 2) Legal privacy controls



4.1. Technical Privacy Controls ⁽¹⁾

- Technical controls - Privacy-Enhancing Technologies (PETs)

[cf. Simone Fischer-Hübner]

a) Protecting user identities via, e.g.:

- **Anonymity** - a user may use a resource or service without disclosing her identity
- **Pseudonymity** - a user acting under a pseudonym may use a resource or service without disclosing his identity
- **Unobservability** - a user may use a resource or service without others being able to observe that the resource or service is being used
- **Unlinkability** - sender and recipient cannot be identified as communicating with each other

- **Taxonomies of pseudonyms** [cf. Simone Fischer-Hübner]
 - Taxonomy of pseudonyms **w.r.t. their function**
 - i) **Personal** pseudonyms
 - Public personal pseudonyms / Nonpublic personal pseudonyms / Private personal pseudonyms
 - ii) **Role** pseudonyms
 - Business pseudonyms / Transaction pseudonyms
 - Taxonomy of pseudonyms **w.r.t. their generation**
 - i) Self-generated pseudonyms
 - ii) Reference pseudonyms
 - iii) Cryptographic pseudonyms
 - iv) One-way pseudonyms

b) Protecting **usee identities** via, e.g.: [cf. Simone Fischer-Hübner]
Depersonalization (anonymization) of data subjects

- **Perfect** depersonalization:
 - Data rendered anonymous in such a way that the data subject is no longer identifiable
- **Practical** depersonalization:
 - The modification of personal data so that the information concerning personal or material circumstances can no longer **or only with a disproportionate amount of time, expense and labor** be attributed to an identified or identifiable individual
- **Controls** for depersonalization include:
 - **Inference controls** for statistical databases
 - **Privacy-preserving methods** for data mining



4.1. Technical Privacy Controls (4)

- The risk of reidentification (a threat to anonymity)

[cf. Simone Fischer-Hübner]

- Types of data in statistical records:
 - Identity data - e.g., name, address, personal number
 - Demographic data - e.g., sex, age, nationality
 - Analysis data - e.g., diseases, habits
- The degree of anonymity of statistical data depends on:
 - Database size
 - The entropy of the demographic data attributes that can serve as supplementary knowledge for an attacker
- The entropy of the demographic data attributes depends on:
 - The number of attributes
 - The number of possible values of each attribute
 - Frequency distribution of the values
 - Dependencies between attributes

c) Protecting confidentiality and integrity of personal data via, e.g.:

[cf. Simone Fischer-Hübner]

- Privacy-enhanced identity management
- Limiting access control
 - Incl. formal privacy models for access control
- Enterprise privacy policies
- Steganography
- Specific tools
 - Incl. P3P (Platform for Privacy Preferences)



4.2. Legal Privacy Controls ⁽¹⁾

- Outline

- a) Legal World Views on Privacy
- b) International Privacy Laws:
 - Comprehensive Privacy Laws
 - Sectoral Privacy Laws
- c) Privacy Law Conflict European Union vs. USA
- d) A Common Approach: Privacy Impact Assessments (PIA)
- e) Observations & Conclusions

a) Legal World Views on Privacy ⁽¹⁾

[cf. A.M. Green, Yale, 2004]

- **General belief:** Privacy is a fundamental human right that has become one of the most important rights of the modern age
- Privacy also recognized and protected by individual countries
 - At a minimum each country has a provision for rights of inviolability of the home and secrecy of communications
 - Definitions of privacy vary according to context and environment

United States: "Privacy is the right to be left alone" -
Justice Louis Brandeis

UK: "the right of an individual to be protected against
intrusion into his personal life or affairs by direct physical
means or by publication of information"

Australia: "Privacy is a basic human right and the reasonable
expectation of every person"

b) International Privacy Laws

[cf. A.M. Green, Yale, 2004]

- Two **types of privacy laws** in various countries:
 - 1) **Comprehensive Laws**
 - Def: General laws that govern the collection, use and dissemination of personal information by public & private sectors
 - Require commissioners or independent enforcement body
 - Difficulty: lack of resources for oversight and enforcement; agencies under government control
 - Examples: European Union, Australia, Canada and the UK
 - 2) **Sectoral Laws**
 - Idea: Avoid general laws, focus on specific sectors instead
 - Advantage: enforcement through a range of mechanisms
 - Disadvantage: each new technology requires new legislation
 - Example: United States

Comprehensive Laws - European Union

- European Union Council adopted the new **Privacy Electronic Communications Directive** [cf. A.M. Green, Yale, 2004]

- Prohibits secondary uses of data without informed consent
- No transfer of data to non EU countries unless there is adequate privacy protection
 - Consequences for the USA

- **EU laws** related to privacy include
 - 1994 — EU Data Protection Act
 - 1998 — EU Data Protection Act
 - Privacy protections stronger than in the U.S.

Government and private databases

❖ Risks

- Unauthorized use by “insiders”
- Inadvertent leakage through negligence, carelessness, hacking
- Propagation of errors
- Intentional: marketing, decision making, surveillance
- Cross-matching files

Threats to Privacy

Threats to Privacy

Information: Government and private databases

- Ministry of the Interior
- Ministry of Housing
- IDF
- National Security
- Pension Funds
- Income Tax
- Banks
- Hospitals and Sick Funds
- Schools & universities

Information: Government and private databases

- Libraries
- Club memberships
- Employers
- Airlines and travel agencies
- Car registration
- Telephone companies
- Census
- ...

Information: Government and private databases

- Individually, most do not pose much threat to privacy
- Collectively, a lot!

Hacking

- ❖ Hacking tips are widely available
- ❖ Threat to privacy
- ❖ Trying different passwords is trivial hacking
 - *use hard-to-crack password!*

Monitoring

- ❖ *Employee Monitoring -*
how much should one tolerate?

Monitoring

"Firms Crack Down on E-Mail"



USA Today (06/28/00) P. 2B; Yaukey, John

Employers are increasingly monitoring workers' email and Internet use due to concerns about liability and lost productivity. Roughly 45 percent of U.S. companies monitor employees' electronic activities such as email, Internet use, and voice mail, and that figure is expected to rise in the future, according to the American Management Association.



Monitoring

Some employees who view monitoring as an invasion of their privacy have taken the issue to court, but judges tend to uphold a company's right to monitor equipment that belongs to the firm.

Privacy

- ❖ Private data are used in matching and profiling to provide
 - Personal/group benefits
 - **Loan - support application**
 - **Job application**
 - Public interest protection
 - **Tax evasion**
 - **Criminal surveillance and identification**

Consumer Information

- ❖ Released through:
 - Marketing survey
 - Making donations
 - Making purchases
 - “Transaction-generated information”
- ❖ Data mining to find out more
- ❖ Often collected without awareness
 - *Invisible information gathering*
 - cookies

Consumer Information

Consumer Databases

Gathering Information:

- Warranty cards.
- Purchasing records.
- Membership lists.
- Web activity.
- Change-of-address forms.
- Much more...

Q: Recall ways in which you have contributed to consumer databases.

Consumer Information

Consumer Databases (cont'd)

Limiting Collection, Use, Sharing, and Sale of Personal Data:

- Consumers can take measures to restrict the use of their personal information.
- Some information sharing is prohibited by law.
- Some information sharing is prohibited by published, privacy policies.

Q: What measures do you take to limit consumer information gathered about you?

Consumer Information

Marketing: Using Consumer Information

- Trading/buying customer lists.
- Telemarketing.
- Data Mining.
- Mass-marketing.
- Web ads.
- Spam (unsolicited e-mail).

Q: How are children affected by marketers using consumer information?

Consumer Information

Credit Bureaus

Uses of consumer information:

- Evaluate credit risk of applicant.
- Marketing.

Regulation:

- FCRA (Fair Credit Reporting Act)
- Self-regulated by privacy principles.

Q: If you are denied credit, what are your rights based on the FCRA?

More Privacy Risks

Social Security Numbers (SSNs)

Appear in:

- Employer records.
- Government databases.
- School records.
- Credit reports.
- Consumer applications.
- Many other databases.

Q: What are the risks of using SSNs as identifiers?

More Privacy Risks

National ID Card System

If implemented, the card could contain your:

- Name.
- Address.
- Telephone number(s).
- Photo.
- SSN.

Q: What other personal information should a national ID card contain?

More Privacy Risks

National ID Card System

If implemented, the system could allow access to your:

- Medical information.
- Tax records.
- Citizenship.
- Credit history.
- Much more...

Q: Are the benefits of a national ID system greater than the risks?

More Privacy Risks

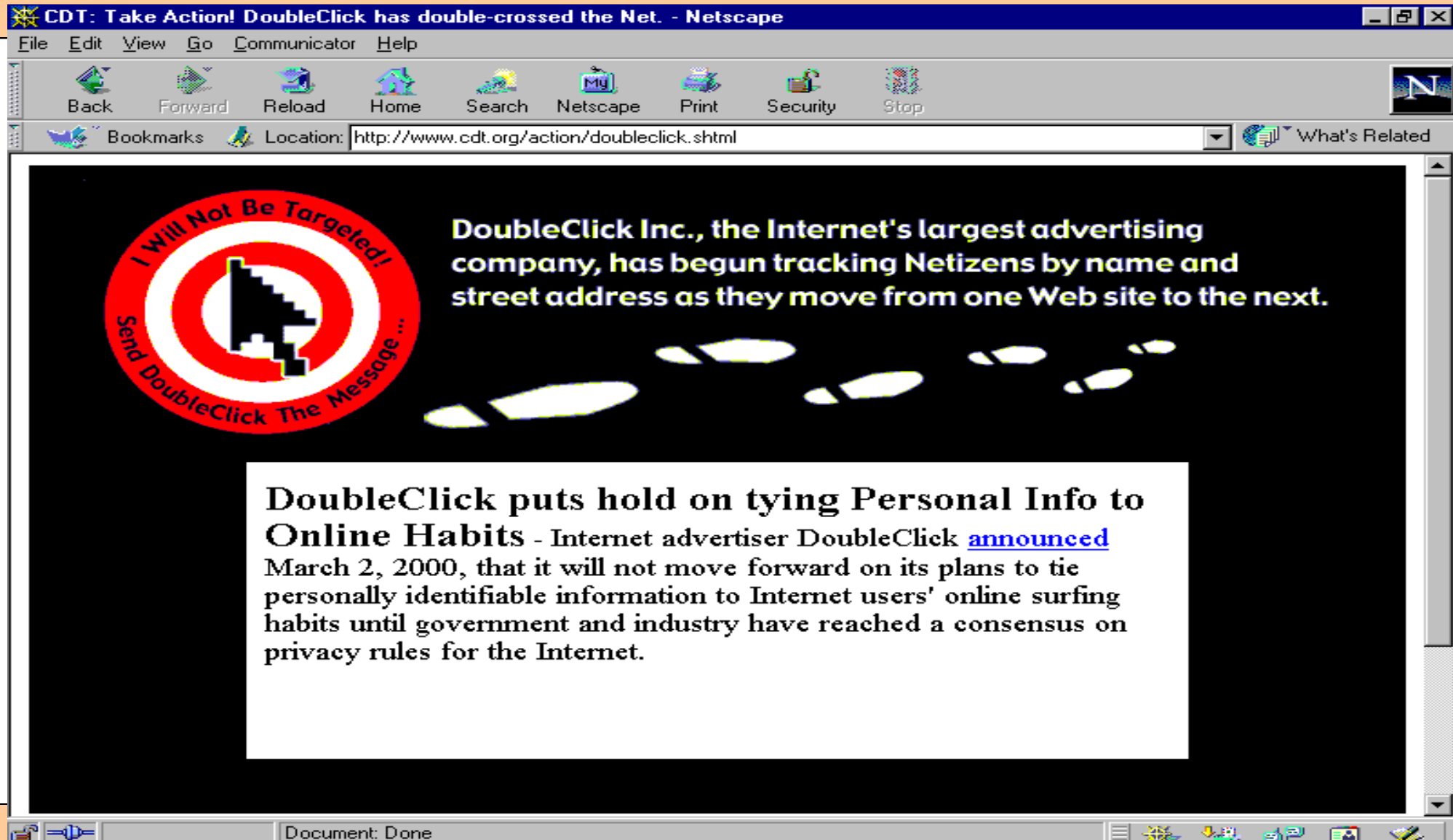
Personal Health and Medical Information

Data can include:

- History of substance abuse.
- Treatment for sexually transmitted disease.
- Extent of psychiatric help received.
- Any suicide attempt(s).
- Diagnosis of diseases (diabetes, angina, cancer, etc.).
- Use of prescribed medicines.
- Much more...

Q: Why would marketers want access to your medical information?

DoubleClick Inc.



DoubleClick Inc.

- ❖ A leading provider of comprehensive global Internet advertising solutions for marketers and Web publishers.
- ❖ Ties individuals' names, addresses, phone numbers, and emails to information collected through cookies (*on-line profiling*) without consent

Consumer Information

- ❖ Result may cause inconvenience, embarrassment and threat
 - e.g. list of consumers who are likely to buy products for adults who are incontinent was made public
 - junk mail

Medical Records

- ❖ How much is protected?
 - ❖ Can insurance company access everything?
 - Can one check on AIDS patients?

Wiretapping

- ❖ Interception of a communication in a telephone or computer network

Wiretapping

- ❖ Telephone bugging now extends to wiretapping on fax transmission, email, voice mail and computer networks
- ❖ Account numbers, passwords, PINs used to be obtainable from touch tones; now they are in emails, or web communications

Privacy

Protecting Privacy

Privacy

- ❖ Three key aspects:
 - Freedom from intrusion
 - Control of information about oneself
 - Freedom from surveillance
- ❖ Privacy is given up for many activities

Privacy

- ❖ Factors to consider:
 - Safeguarding individual and group privacy
 - Collecting information for rational decision making
 - Conducting surveillance within laws to protect public order and safety

Encryption

- ❖ Cryptography – “the art and science of hiding data in plain sight”
- ❖ To transform a message or data into a form that is meaningless to anyone who might intercept it.

Encryption

❖ Uses:

- Electronic transfer of funds
- Passwords and PINs for consumer transaction
- Credit card numbers
- Bank records and financial data
- Sensitive business communications
- Research and product information
- Personal data and communications
- Test results

Encryption

- ❖ Cyphertext – the coded message
- ❖ Decryption – the decoding process
- ❖ Key – a specific sequence of characters for coding/decoding

Encryption

❖ Example of an encryption key:

qwertyuiopasdfghjklzxcvbnm

- “a” becomes “q”
- “b” becomes “w”
- Etc.

Public Key Cryptography

- ❖ Beforehand, key must be known to both parties
- ❖ Public key and private key
 - Encrypting key is made public
 - Public key (of the recipient) is used to encrypt message sent
 - Private key is used to decrypt by the recipient

PGP (Pretty Good Privacy)

- ❖ A program developed by Philip Zimmermann using public key cryptography for emails
- ❖ Provides protection of emails from wiretapping
- ❖ Widely and freely available

PGP (Pretty Good Privacy)



Welcome to the MIT Distribution Center
for PGP (Pretty Good Privacy)

<http://web.mit.edu/network/pgp.html>

Digital Signatures

- ❖ Some public key cryptography schemes allow the reversal of public and private keys to provide digital signatures
 - A person accepts an electronic document and encrypts it with his private key. Recipient can then decrypt the document with the public key.

Privacy Protection

- ❖ File wiping
 - simple 'del' or 'rm' removes only file name from index; trash leaves it in place until emptied
 - random over-writing with '0' and '1' provides better protection
 - many 'wipe' utilities exist
- ❖ Empty cache

Privacy Protection

- ❖ Choosing a good password
 - Never use common phrases
 - Randomize
 - Change frequently
- ❖ Forgotten your password
 - many websites offer services to recover lost data

Privacy Protection

- ❖ Get a separate email account for personal email
- ❖ Clear your memory cache after browsing
- ❖ Make sure that online forms are secure
- ❖ Reject unnecessary cookies
- ❖ Opt-out of third party information sharing
- ❖ Use anonymous remailers

Three Dilemmas

Three Dilemmas

Maintaining Privacy

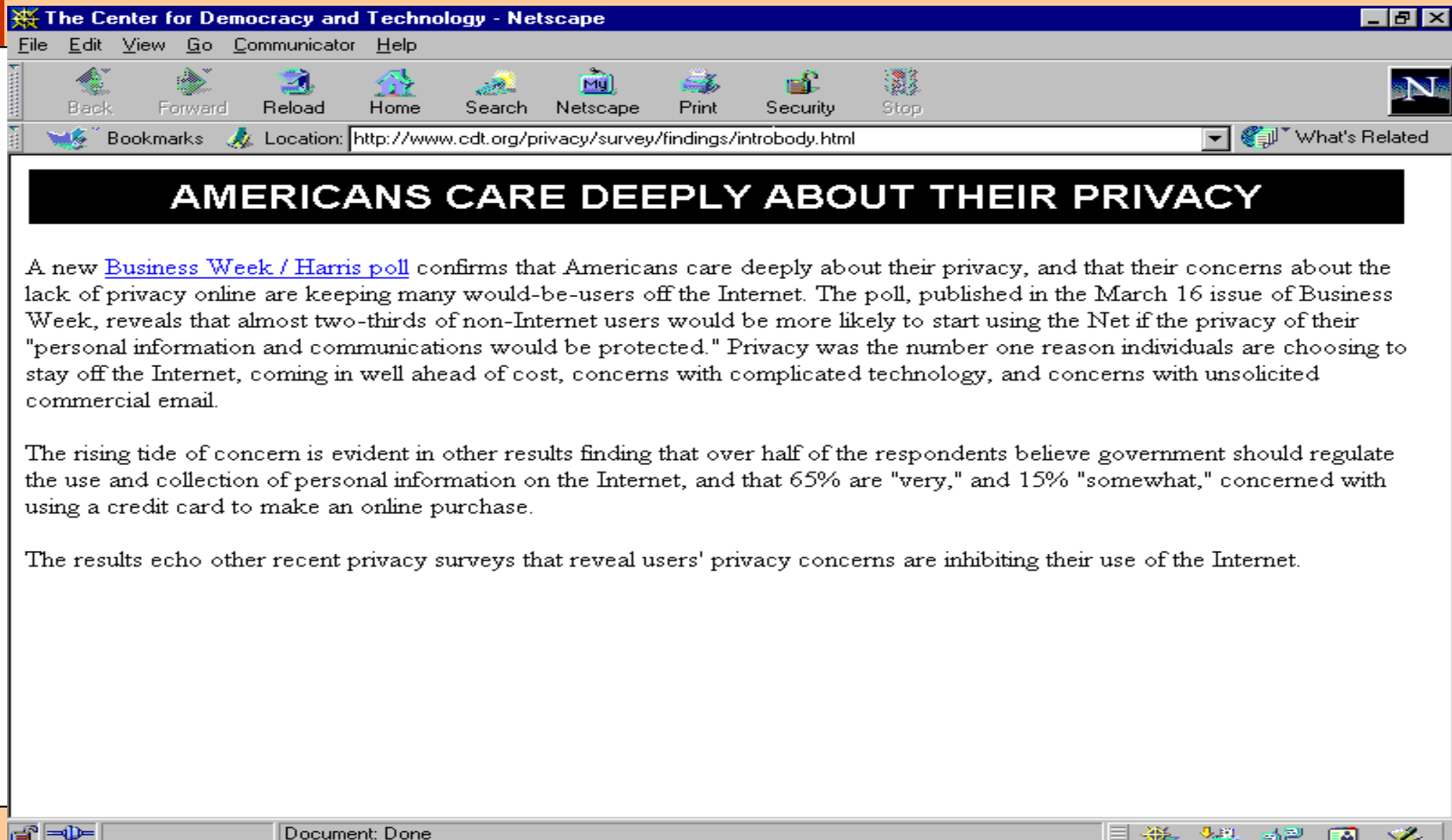
❖ *Three Dilemmas*

- For individuals: how to obtain good service without compromising their privacy?
- For businesses: how to provide good service without offending customers?
- For governments: how to prevent misdeeds without compromising individual privacy?

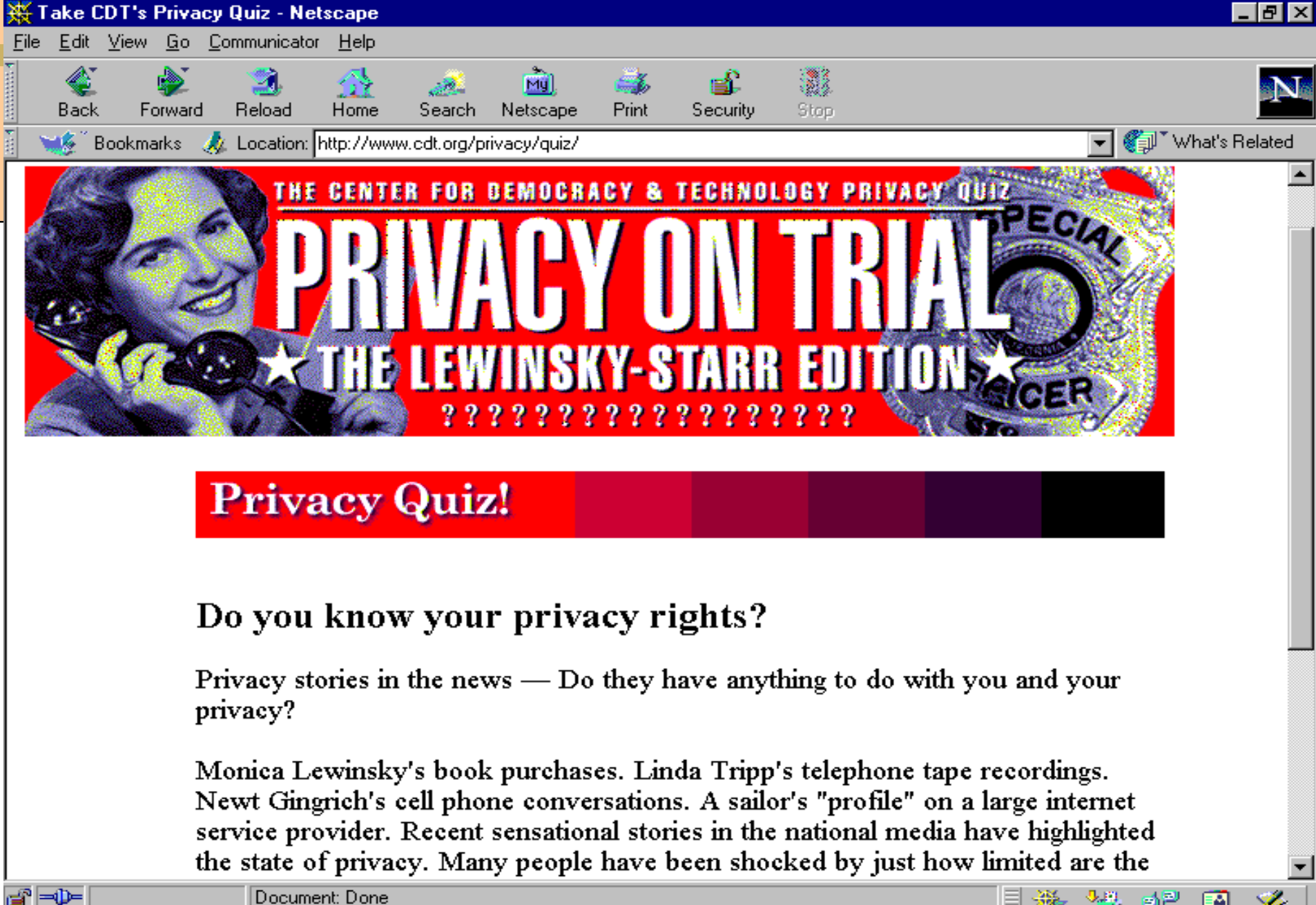
Individuals

Individuals & Privacy

Privacy the reason for 'off-line'?







<http://www.cdt.org/privacy/quiz/>

Anonymity and Pseudonymity

- ❖ Hide identity for self-protection:
 - voice out against government
 - voice out against employer
 - “off” the record
- ❖ Use pseudonymous remailers
- ❖ Programs available for anonymous Web browsing, email encryption, and chat room anonymity

Anonymity and Pseudonymity

Allowing
anonymity/
pseudonymity = *Free
Speech* ?

Individuals

latimes.com

August 21, 2000

Many Online Taking Privacy Into Their Own Hands

By JUBE SHIVER JR., Times Staff Writer

WASHINGTON--The nation's Internet users want stronger privacy protections in cyberspace and many resort to using fake names, dummy e-mail accounts and data scrambling software to preserve their anonymity, a new study has found.

Individuals

latimes.com

The study, by the Pew Research Foundation in Washington, reported that 84% of Internet users said they are concerned about businesses invading their privacy online. By that count, commercial Web sites outranked even hackers, considered a privacy threat by only 68% of those surveyed.

"There is broad-based concern about privacy being compromised," said Lee Rainie, the director of the study. Americans, Rainie added, "'want the golden rule of the Internet to be: 'Don't do anything unto me unless I give you permission.' "

Individuals

latimes.com

Yet even as they express anxiety about being monitored online, consumers' ability to protect their privacy is hampered by their unfamiliarity with the basic mechanics of Internet data collection.

For example, 56% of Internet users surveyed said they did not know what an Internet "cookie" is. So-called cookie files are unique identifiers many Web sites place on visitors' computers in order to be able to track their online movements.

Individuals

latimes.com

Many computer users--even those who buy goods online--are simply "unaware that their computers' hard drives are implanted with cookies," the report said.

The study also noted that despite their privacy worries, many Web surfers do a striking number of intimate and trusting things online, such as responding to e-mail from strangers or making their personal address books or appointment schedules accessible online.

Organizations

Organizations & Privacy

Organizations

The newest high-level position: Chief Privacy Officer
Companies are hiring them to ease consumers' concerns in this age of the Internet - and to prevent costly suits.

By D. Ian Hopper

ASSOCIATED PRESS

With consumers increasingly concerned about their privacy, and new technology able to track Internet users click by click, companies are rapidly hiring privacy officers, and giving them broad powers to set policies that protect consumers from invasion and companies from public-relations nightmares.

In many cases, the privacy officers report directly to the chairman or chief executive officer. And their hiring has become a litmus test for a company's dedication to customer privacy.

Organizations

Big Brother is here, consumers say

In addition to expressing concerns about privacy on e-business sites, consumers are worried about being watched by the government.



*Mean scores are based on an 11 point scale where 0 means strongly disagree and 10 means strongly agree

SOURCE: CHESKIN GROUP

Organizations

REUTERS  Technology

Web advertisers strike privacy deal

By Andy Sullivan, 28 July 2000

A group of Internet advertisers announced Thursday a new set of industry standards crafted with the federal government to give Web surfers a say in how their personal data is used by online marketing firms.

The deal also bars Internet firms from using visitors' medical or financial data, Social Security numbers and online sexual behaviour to determine which advertisements to flash on their screens.

Organizations

REUTERS  Technology

The FTC issued a report that praised the NAI deal but also called on Congress to pass a consumer-privacy law to cover firms the organization does not represent.

"Self-regulation is an important and powerful mechanism for protecting consumers, and the NAI principles present a solid self-regulatory scheme," the FTC said. "Nonetheless, backstop legislation addressing online profiling is still required to fully ensure that consumers' privacy is protected online."

Interest in consumer privacy on the Internet is growing in Congress. Yesterday, Arizona Sen. John McCain, the powerful chairman of the Senate Commerce Committee, introduced a bill that contained many of the provisions in the agreement.

Organizations

July 3, 2000

Not Enough Privacy?



European legislators are about to vote on an online privacy deal with the United States. Guess where people will get more protection.

By Keith Perine

Concerned about internet privacy? You might want to move to Europe. The European Parliament was scheduled to vote July 4 on a proposed agreement between the United States and the 15-nation European Union that would grant Europeans greater online protection from U.S. companies than they legally are required to provide to Americans.

Organizations



Under the terms of the agreement, U.S. companies that want to transfer personal data on Europeans must commit to detailed standards of notice, user choice, data access and security. Doing so would put the firms into "safe harbor" against regulation under the EU's omnibus Data Protection Directive.

"The Europeans have always stood their ground and demanded their privacy, and because of that the safe harbor principles improved enormously," says privacy advocate Jason Catlett, president of Junkbusters. "Americans are starting to ask, 'Why are Europeans getting better privacy protection than we are?'"

Organizations



The provisions of the EU law require businesses to collect private data only for clearly stated purposes and forbid data disclosure to third parties unless consumers grant permission. European consumers have the right to sue companies that don't adhere to the rules. European subsidiaries of U.S. companies already have to abide by the directive, which is enforced in many EU countries by government data commissioners.

The FTC used to champion pure industry self-regulation, as has the Commerce Department. But earlier this year, the commission changed course. While its first two annual online privacy reports to legislators in 1998 and 1999 touted pure self-regulation, this year's report recommended federal legislation.

Governments

Governments & Privacy

Governments



Grim Net Censorship Report

Reuters

6:00 a.m. Apr. 27, 2000 PDT

New York -- Censorship of the Internet by governments is spreading and may become a threat to traditional media liberty, a report on press freedom said on Wednesday.

"The explosion of news and information on the World Wide Web is tempting governments, developed and developing, politically free and not free, to consider restricting content on the Internet," said the report conducted by human rights group Freedom House.

Governments

The New York Times
ON THE WEB

July 19, 2000

British Authorities May Get Wide Power to Decode E-Mail

By SARAH LYALL

London: As the Clinton administration formally enters the debate about law enforcement surveillance in cyberspace, the British government is about to enact a law that would give the authorities here broad powers to intercept and decode

A necessary tool for fighting Internet crime or a violation of civil liberties?

e-mail messages and other communications between companies, organizations and individuals.

Governments

The New York Times
ON THE WEB

The measure, which goes further than the American plan unveiled on Monday in Washington, would make Britain the only Western democracy where the government could require anyone using the Internet to turn over the keys to decoding e-mails messages and other data.

Government officials maintain that the measure is essential if law enforcement agencies are to combat the sophisticated modern crime that is enhanced by access to the Internet, including pedophilia, drug smuggling, money laundering, terrorism and trafficking in refugees.

Governments

The New York Times
ON THE WEB

But the measure has had a rocky time in Parliament, where lawmakers have vehemently objected to several provisions, including one that would give the government new powers to require Internet service providers to install "black box" surveillance systems that would sort and send a range of data and e-mail to a monitoring center controlled by the domestic security service, M.I.5.

Such systems are also being used in the United States by the Federal Bureau of Investigation, where the technology is known as Carnivore because it is able to extract the "meat" quickly from vast quantities of e-mail messages and other communications between computers.

Governments

The New York Times
ON THE WEB

In contrast to the United States, Britain has a tradition of unfettered and often uncontested intrusion by the authorities into citizens' privacy.

In the United States, the F.B.I. must first obtain a search warrant before using the Carnivore technology, which is then installed and maintained by the bureau.

Under the British plan, failure to turn over a decryption key or to convert encrypted data or messages into plain text could result in a two-year prison sentence. Although many nations are considering similar bills to deal with encrypted data, only Singapore and Malaysia have so far enacted them.

A necessary tool for fighting Internet crime or a violation of civil liberties?

Governments



Congress Takes on Internet Privacy Legislation

By Andy Sullivan

Updated 3:12 PM ET January 23, 2001

WASHINGTON (Reuters) - Two U.S. lawmakers unveiled legislation on Tuesday providing some privacy protection for Internet users, adding yet another approach to what will likely be a major issue for the 107th Congress.

The bill ... Would require Web sites to notify visitors how personal data such as telephone numbers and ZIP codes are used, and allow visitors to limit its use.

Governments



Today's Cartoon

By Jerry King

Special to the E-Commerce Times

July 27, 2000



"We know you have a crush on Jimmy. The FBI forwarded us the e-mail."

Governments

From the editor-in-Chief| Michael Vizard
"Should We Really Be Asking the Federal Fox to Guard
the Online Privacy Henhouse?"

InfoWorld (08/07/00") Vol. 22, No. 32,



... now may be the time to form an independent government body to create and enforce a privacy policy for the Internet. After all, allowing the government to create a privacy policy for the Internet when it has a vested interest in violating it is like asking a fox to guard a henhouse.

Governments

The government has already spent years trying to limit encryption technologies so it can track criminals who use the technology. The government has also developed Carnivore technology so it can track an individual's email messages across multiple Internet service providers. "Clearly, there is going to be a requirement to give individuals some reasonable expectation of privacy on the Internet," writes Vizard. Having Congress create an impartial organization exclusively dedicated to setting privacy policy would be the challenge.

Governments



The head of the privacy body would be nominated by the president and confirmed by Congress for a six-year term. Privacy, a commodity more precious than money, deserves such standards, Vizard contends.

<http://www.infoworld.com/articles/op/xml/00/08/07/000807opvizard.xml>

Governments

InternetNews - Business News

July 17, 2000

U.S. Legal Body: Internet May Need New Cyber-Borders

The Internet makes such light work of geographical frontiers that new cyber-borders may be needed instead, top U.S. lawyers said on Monday as they presented a two-year report into preventing global online chaos.

Enthusiasts may love the Internet's scant regard for authority and borders but it presents a major headache for business, government and also consumers. a French customer buys a rug from Turkey via a website hosted in the United States and with a Swiss credit card, for example, there are risks all round -- the rug might be a dud, the payment might be faulty and taxes might not be paid -- but where should such matters be settled?

Privacy Policy

Privacy Policy

Privacy Policy

ELEMENTS of PRIVACY POLICIES

**Karen A. Forcht and Malik Ali
James Madison University**

Organizations establishing privacy policies should incorporate the elements of the widely accepted Code of Fair Information Practice, which states that:

Privacy Policy

- **There must be a way for an individual to find out what information about him or her is in a record and how it is used.**
- **There must be a way for an individual to correct or amend a record with information that is identifiable to him or her.**
- **The existence of all data systems with personal information in them should be publicly disclosed, and the purpose for which information is gathered about people should be disclosed.**

Privacy Policy

- **There must be a way for an individual to prevent information about him or her that was obtained for one purpose (which was stated when the information was gathered) from being used or made available, either within the organization or outside, for a purpose that is incompatible with the original purpose, without getting the consent of the individual. This is the principle of secondary use.**

Privacy Policy

- **The organization creating, maintaining, using, or disseminating records of identifiable personal data must assure the reliability, accuracy, security and timeliness of the data. In other words, the custodian of information that is disseminated has an obligation to the individual to make sure it is accurate, secure, and not misused. This obligation ought not be delegated to another entity**

Privacy Policy

- **An organization must conduct periodic risk assessments, balancing the possibility or probability of unauthorized access or disclosure against the cost of security precautions and the expected effectiveness of the precautions.**
- **An organization must make sure that other entities handling personal information in behalf of the first organization are bound by these same principles.**

Privacy Policy

- Organizations must take special precautions in collecting and using personal information about children for those 18 and younger.
- An organization should openly disclose its policies and practices with regard to electronic surveillance of its employees' and customers' telephone calls, electronic mail, Internet usage, changing rooms, and rest rooms, and must articulate in advance the reasons for the surveillance.

Agenda

- **Computers and privacy**
- **Threats to privacy**
- **Protecting privacy**
- **Three dilemmas**
 - **Individuals and privacy**
 - **Organizations and privacy**
 - **Governments and privacy**
- **Privacy policy**

Database Nation

- **Tiny chunk of data can reveal your home address**
- **Linking to People searches can reveal your phone number**
- **Your county government's Web displays your home's floor plan and assessed value**
- **Pay-as-you-go databases like Lexis-Nexis' P-TRAK, P-FIND, and P-SEEK tie together mortgage records, vehicle registrations, court judgments, bankruptcy histories, and any other public information they can gather.**

Database Nation (contd)

- **Google and Yahoo! can record every search you've ever made and link it to whatever computer you used at the time.**
- **Credit card companies know what you buy**
- **frequent shopper programs know what you eat.**
- **Your insurance company knows what medical procedures you've undergone.**

Database Nation (contd)

- **In a February 2003 Harris poll, 69 percent of those surveyed agreed that "consumers have lost all control over how personal information is collected and used by companies."**
- **Sun Microsystems CEO Scott McNealy. "You have zero privacy anyway," he said a few years ago. "Get over it."**

Privacy Protection Under the Law



Privacy Protection Under the Law

While currently there is no national law to protect the privacy of the information you share online, federal law and state law do offer some protection to various kinds of personal information collected about you.

Privacy Protection Under the Law

- The Fair Credit Reporting Act (1970)
- Privacy Act of 1974
- Family Education Rights and Privacy Act (1974)
- Right to Financial Privacy Act (1978)
- Privacy Protection Act of 1980
- Cable Communications Policy Act of 1984
- Electronic Communications Privacy Act (1986)
- Video Privacy Protection Act of 1988
- Telephone Consumer Protection Act of 1991
- Driver's Privacy Protection Act of 1994
- Communications Assistance for Law Enforcement Act of 1994
- Telecommunications Act of 1996
- Health Insurance Portability and Accountability Act of 1996
- Children's Online Privacy Protection Act (COPPA) of 1998
- Financial Modernization Act (Gramm-Leach-Bliley Act) (2000)

Privacy Policy



Privacy Policy

- **What information is being collected?**
- **How long is personal information stored?**
- **How is personal information used once it is collected?**
- **Will this information be shared with others?**
- **Can users access information that has been collected about them?**
- **What laws govern the collection?**

Privacy in the news

Sony BMG Violated the Children's Online Privacy Protection Act

(December 11, 2008)

- **agreed to pay \$1 million as part of a settlement**
- **The company operates over 1,000 Web sites for its musical artists and labels**
- **On 196 of these sites, Sony Music knowingly collected personal information from at least 30,000 underage children without first obtaining their parents' consent,**
- **largest penalty ever in a COPPA case.**

Privacy in the news

Mortgage Company Settles Data Security Charges **(December 11, 2008)**

- **Premier Capital Lending, Inc. (Premier) violated the FTC's Safeguards and Privacy Rules, as well as Section 5 of the FTC Act.**
- **Were required to implement reasonable policies and procedures to ensure the security and confidentiality of sensitive customer information (Gramm-Leach-Bliley Act,)**
- **400 credit reports through Premier's account were stolen.**

Privacy in the news

Settlement against Retailer TJX

(March 27, 2008)

- **2,500 stores worldwide.**
- **failed to use reasonable and appropriate security measures to protect customer information**
- **Personal information of the personal information of approximately 455,000 consumers was stolen.**

What can you do about it?



Conclusion

As a business owner

- Stay trustworthy to your customers.
- Ask what you need.
- Protect what you promise

As a consumer

- Look for and read privacy policies on the Web.
- Talk to your children about getting your permission before giving out information online.
- Opt-out of third party information sharing.
- Don't give companies more information than you have to.
- and most importantly...



SeaNSun LLC

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Are you giving your privacy away?

The Google logo, consisting of the word "Google" in its signature multi-colored font.

71% of people don't know that some email services scan the text of the messages they send and receive.

The Yahoo! logo, featuring the word "YAHOO!" in a bold, red, sans-serif font.

87% feel that using the content of their messages in this way is an invasion of their privacy.

The Bing logo, featuring the word "bing" in a blue, lowercase, sans-serif font with a small orange dot above the "i".

Data from a survey by Mozaic group

Privacy

- Although email is a revolutionary invention, it still has some flaws.



Privacy

- Most emails are not encrypted, so can be hacked
- Email has to go through another computer to reach you, so it may be intercepted and read



Privacy

- Your internet service provider stores copies of emails you have received which may be read if required permission
- Headers and forwarded messages can reveal your email address to anyone that receives the message



Privacy

- So, in fact, email is not very 'private' at all.



Invasion of Privacy?



Microsoft lays into Google for 'invading Gmail users privacy' by scanning contents of emails to target personalised adverts

- Latest attack is part of the software company's Scroogled campaign
- Includes sinister quotes from Google executive chairman Eric Schmidt
- But Microsoft's privacy policy also allows it to serve personalised ads

By DAMIEN GAYLE

PUBLISHED: 11:46 EST, 8 February 2013 | UPDATED: 12:28 EST, 8 February 2013

These email service providers launch marketing campaigns accusing each other of invading users privacy.



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Free Email!



They are happy to give email away for free, the data they collect is worth millions.



Their free email comes with a price, your privacy!



Some even charge for advanced features, upgrade now for the privilege of paying them to invade your privacy.

Your mail and so much more

Get a whole year for \$19.99—that's less than \$2/month!

Upgrade Now



SeaNSun LLC

Small Business Solutions

www.seansunllc.com

Invade privacy, even non-users



Public opinion, at least in the U.S., is very much against this practice. Nearly 70 % don't even know it's happening and when they do find out, 88 % say it should stop.

No need for a warrant - requests

Google gives user data to government in most cases

0 Comments





NEW Portfolio Relevance
LEARN MORE

By John Letzing, MarketWatch

SAN FRANCISCO (MarketWatch) — Google Inc. disclosed on Monday that it complied with 94% of government requests for data about users of its services in the U.S. in the second half of last year, shedding some light on the Internet search firm's behind-the-scenes interaction with law enforcement.

WHAT IS WEB PRIVACY?

- Web privacy is the concept of how to maintain certain standards of control over information procured, used and stored on the Internet.
- In other words, web privacy keeps data about a person hidden when accessing websites or simply surfing the web.



PROBLEMS

- How to provide personalization of Internet services while still protecting users' privacy
- → when we *read information or chat on a web site*, how can we ensure that our privacy is preserved?
- → when we *search for information with a search engine*, how can we ensure that traces of our searches will not be used for purposes outside our control?
- → when we *register on a website*, how can we ensure that private data will not be disseminated on the web to other parties we do not trust?



WHAT'S THEIR PROFIT???

- Why are the companies worth so much money?
- Why is MySpace, Facebook, Flickr worth over half a billion dollars without a proper revenue model?



HOW 'THEY' CAN INFILTRATE OUR PRIVACY?

- Information we gather and track
- Personal information
- Aggregated Tracking Information
- Cookies
- Third party cookies
- Third-party advertising



WEB 2.0 SECURITY' S RISK LEAD TO PRIVACY VULNERABILITIES

- **Insufficient Authentication Controls**
- **Cross Site Scripting (XSS)**
- **Information Leakage**
- **Cross Site Request Forgery**
- **Fishing**



RECOMMENDATIONS

- **Private information**
- **Clean out your search history**
- **Clean up your cookies**
- **Clean out your cache**
- **Search with a proxy**
- **Avoid registrations**
- **Disable auto-complete**
- **Careful of your downloads**



Offensive Speech and Censorship in Cyberspace

Speech Might Include:

- Political or religious speech.
- Pornography (Multibillion-dollar online business).
- Sexual or racial slurs.
- Nazi materials.
- Libelous statements.
- Abortion information.
- Alcohol ads.

Q: Identify other forms of speech found in cyberspace that some consider offensive.

Offensive Speech and Censorship in Cyberspace

Miller v. California

Material is considered obscene if all three parts are met:

1. It depicts sexual (or excretory) acts whose depiction is specifically prohibited by state law, and
2. It depicts these acts in a patently offensive manner, appealing to the prurient interest as judged by a reasonable person using community standards, and
3. It has no serious literary, artistic, social, political, or scientific value.

Q: Historically, how have local “community standards” affected censorship of speech in cyberspace? (see next page)

Offensive Speech and Censorship in Cyberspace

Material Inappropriate for Children

Technology Changes the Context

- On the Web, children have access to the same 'adult' text, images, videos, etc. as adults (no "tickettaker" to check age).
- Online proprietors don't know the customer is not an adult.

Protecting Children

Regardless of the medium:

- It is illegal to create, possess or distribute child pornography. (Because creation of it is child abuse...)
- It is illegal to lure children into sexual activity.

Q: How should children be protected from access in cyberspace to adult material? (next page)

Protect Children

Software filtering:

Bess

Cyber Patrol, Surf-Watch

Offensive Speech and Censorship in Cyberspace

Censorship Laws

Communications Decency Act (CDA, 1996)

- Publicity and public pressure lead Congress to pass this act.
- Anyone who made available to anyone under 18 any communication that is obscene or indecent would be subject to a \$100,000 fine and two years in prison.
- American Civil Liberties Union (and other free speech advocates) found this law to be a “profound threat to freedom of expression.
- In 1997, the CDA was ruled unconstitutional because it was too vague and too broad in protecting children online and because less restrictive means are available.

Q: Should content on the Internet have as much First Amendment protection as printed material?

Offensive Speech and Censorship in Cyberspace

Censorship Laws (cont'd)

Child Online Protection Act (COPA, 1998)

- Commercial Web sites that make available to minors materials “harmful to minors”, as judged by community standards would be subject to a \$50,000 fine and six months in jail.
- In 2000 and 2003, COPA was ruled unconstitutional by a federal court.

Q: How are children protected from “harmful” material outside of Cyberspace?

Offensive Speech and Censorship in Cyberspace

Censorship Laws (cont'd)

Children's Internet Protection Act (CIPA)

- Any school or library receiving federal Internet funds must install filtering software on all Internet terminals.
- Filters must block sites containing child pornography, obscene material, and any material deemed “harmful to minors.”
- A federal appeals court ruled a major part of CIPA unconstitutional in 2002 but the Supreme Court upheld the law in 2003.

Q: How does CIPA affect adults accessing online material at a public library that uses filtering software? See page 206...

Offensive Speech and Censorship in Cyberspace

Limiting Internet Access in Libraries and Schools

Filtering Software

- Benefit: prevent access to inappropriate material on the Internet by screening words or phrases, blocking sites according to rating system, or disallowing access to specific sites in a list.
- Problems: can be ineffective—kids get around the filters; the words, phrases, rating systems, etc. are subjective; “banned” keywords can be overly restrictive for adult users and for legitimate use by minors.
- See page 205 for a list of filtered sites...

Q: What has been your experience with filtering software?

Offensive Speech and Censorship in Cyberspace

Challenging Old Regulatory Paradigms and Special Interests

License required:

- To practice law.
- To publish traditional newsletters about commodities and futures investing (prior to 2000).

No license required:

- To create downloadable, self-help legal software.
- To publish newsletters about, developing software for, and operating Web sites concerning commodities and futures investing (since 2000).

Q: Is online advertising of wine free speech or disregard for the regulatory rights of state governments?

Offensive Speech and Censorship in Cyberspace

Censorship On the Global Net

Global Impact

- Avoiding censorship: the global nature of the Net allows restrictions (or barriers) in one country to be circumvented by using networks in other, less restrictive countries.
- Creating censorship: the global nature of the Net makes it easier for one nation to impose restrictive standards on others. Cases page 210.

Q: Why are online gambling sites established offshore?

Anonymity

Common Sense and the Internet

- Early publications by some of our Founding Fathers were published under pseudonyms.
- Today, there are publications on the Net that are posted anonymously.
- Anonymizer.com Surf the web/send e-mail anonymously

Q: What are the drawbacks of anonymous Web postings?
(See next slide)

Against Anonymity

- Potential to shield criminal/anti-social activity
 - Fraud, harassment, and extortion
 - Distribution of child pornography
 - Libel or threatening others with impunity
 - Infringe copyrights by posting software
 - Plan terrorist attacks
- Poor “netiquette”

Anonymity

Is Anonymity Protected?

Conflicts between political freedom of speech and campaign regulations:

- Anonymity protects against retaliation and embarrassment.
- Anonymity violates rules established by the Federal Elections Commission (FEC) Case page 215.

Q: Should anonymous political speech on the Web be regulated?

Anonymity

Anonymity vs. Community

Supporters of anonymity:

- Say it is necessary to protect privacy and free speech.

Opponents of anonymity:

- Believe it is anti-social and allows criminals to hide from law enforcement.

Q: How is the practice of anonymity online similar/dissimilar to strong encryption? VERY similar...challenging to law enforcement.

Spam

What Is the Problem?

Unsolicited, mass e-mail:

- is cheap to senders but may impose costs on the recipient's time and/or the recipient's online account.
- may contain objectionable content (political, commercial ads, solicitations for funds, pornography, etc.).
- may contain a disguised return address.
- may pass through filters.
- invades privacy.
- creates a financial and managerial burden on ISPs.

Q: How do you handle spam?

Cases and Free Speech Issues

AOL v. Cyber Promotions

- AOL and other service providers have successfully sued spammers because of the cost burden imposed. Page 219.

Disgruntled Intel Employee

- Initially, a court ruled that non-commercial spam to Intel employees at their Intel e-mail accounts was a form of trespass.
- The CA Supreme Court ruled that it was not.

Spam

Solutions

- Technology: filters that screen out spam.
- Market Pressure: services that list spammers.
- Business Policy: at the discretion of the recipient, all e-mail would be charged a microfee.
- Law: create restrictions that are consistent with the First Amendment.
- Vigilantism: punish spammers by hacking into their phone or computer systems.

Q: Which solution above, or others, do you support?